

## YANGDOK, Korea, Democratic People's Republic of

WMO: 470520

Lat: 39.217N

Lon: 126.650E

Elev: 279

StdP: 98.02

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-19.8 | (c)<br>-17.7 | (d)<br>-23.2                  | (e)<br>0.5 | (f)<br>-17.4 | (g)<br>-21.0 | (h)<br>0.6 | (i)<br>-15.8 | (j)<br>8.4            | (k)<br>-9.4 | (l)<br>7.7 | (m)<br>-8.8 | (n)<br>0.3            | (o)<br>320 | (p)<br>0.423 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>7.4             | (c)<br>30.9     | (d)<br>23.8 | (e)<br>29.6 | (f)<br>23.2 | (g)<br>28.3 | (h)<br>22.4 | (i)<br>25.6         | (j)<br>29.2 | (k)<br>24.8 | (l)<br>28.0 | (m)<br>24.1 | (n)<br>26.8 | (o)<br>2.4           | (p)<br>320 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.5                            | 20.2 | 27.7 | 23.8 | 19.3 | 26.7 | 23.2 | 18.5 | 26.0 | 80.6          | 29.3 | 77.2 | 28.1 | 74.1 | 26.8 | 29.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| Min               | Max  |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 7.8               | 6.8  | 5.9 | -22.8                      | 33.3 | 3.2                | 1.4 | -25.1                                              | 34.3 | -27.0      | 35.1 | -28.8      | 35.9 | -31.2      | 36.9 |  |  |
|                   |      |     | -23.2                      | 27.2 | 2.9                | 1.1 | -25.3                                              | 28.0 | -27.0      | 28.6 | -28.6      | 29.3 | -30.7      | 30.1 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.0    | -7.7  | -4.8  | 1.9   | 8.8   | 15.4  | 20.1  | 23.4  | 23.5  | 18.1  | 11.1  | 2.7   | -5.3 | (6)  |
| (7)  |                                                                                  | DBStd       | 11.56  | 4.68  | 4.54  | 3.97  | 3.87  | 3.01  | 2.26  | 2.50  | 2.46  | 2.80  | 3.67  | 4.88  | 5.03 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2011   | 548   | 413   | 253   | 66    | 2     | 0     | 0     | 0     | 0     | 32    | 222   | 474  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3823   | 807   | 647   | 511   | 286   | 99    | 8     | 1     | 1     | 37    | 226   | 469   | 732  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1651   | 0     | 0     | 1     | 31    | 168   | 304   | 417   | 419   | 243   | 64    | 3     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 422    | 0     | 0     | 0     | 0     | 6     | 63    | 159   | 163   | 30    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 2976   | 0     | 0     | 0     | 11    | 169   | 507   | 1058  | 1055  | 172   | 5     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 720    | 0     | 0     | 0     | 1     | 23    | 116   | 283   | 284   | 13    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 1.7    | 1.5   | 1.9   | 2.1   | 2.2   | 2.2   | 1.7   | 1.5   | 1.4   | 1.4   | 1.5   | 1.8   | 1.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1107   | 13    | 18    | 26    | 56    | 90    | 122   | 317   | 262   | 112   | 52    | 42    | 22   | (15) |
| (16) |                                                                                  | PrecMax     | 1830   | 33    | 66    | 114   | 118   | 201   | 295   | 1034  | 891   | 316   | 162   | 120   | 65   | (16) |
| (17) |                                                                                  | PrecMin     | 593    | 1     | 1     | 2     | 4     | 22    | 31    | 53    | 25    | 10    | 5     | 6     | 2    | (17) |
| (18) |                                                                                  | PrecStd     | 298    | 9     | 15    | 25    | 31    | 47    | 69    | 176   | 191   | 82    | 39    | 29    | 14   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 5.1   | 9.7   | 17.6  | 25.0  | 28.9  | 31.4  | 33.1  | 32.7  | 28.2  | 23.8  | 17.0  | 8.0  | (19) |
| (20) |                                                                                  |             | MCWB   | 2.2   | 6.1   | 9.7   | 14.2  | 16.6  | 20.1  | 24.8  | 25.3  | 21.2  | 15.9  | 12.0  | 5.7  | (20) |
| (21) |                                                                                  | 2%          | DB     | 2.9   | 6.3   | 14.4  | 22.0  | 26.7  | 29.5  | 31.0  | 30.9  | 26.6  | 21.4  | 14.1  | 5.7  | (21) |
| (22) |                                                                                  |             | MCWB   | 0.6   | 2.5   | 8.1   | 12.3  | 16.3  | 19.8  | 24.3  | 25.1  | 19.5  | 15.1  | 10.4  | 3.6  | (22) |
| (23) |                                                                                  | 5%          | DB     | 1.6   | 4.2   | 11.7  | 19.8  | 25.2  | 27.9  | 29.5  | 29.6  | 25.4  | 19.6  | 12.2  | 3.7  | (23) |
| (24) |                                                                                  |             | MCWB   | -0.4  | 1.2   | 6.3   | 11.7  | 15.4  | 19.3  | 23.8  | 24.3  | 18.7  | 13.9  | 9.2   | 2.0  | (24) |
| (25) |                                                                                  | 10%         | DB     | 0.3   | 2.4   | 9.2   | 17.4  | 23.3  | 26.3  | 28.1  | 28.2  | 24.1  | 17.8  | 10.3  | 2.0  | (25) |
| (26) |                                                                                  |             | MCWB   | -1.3  | 0.1   | 4.6   | 10.0  | 14.7  | 18.8  | 23.4  | 23.5  | 18.1  | 12.9  | 7.7   | 0.5  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 2.6   | 7.2   | 11.6  | 16.4  | 19.7  | 22.7  | 26.6  | 26.8  | 23.1  | 17.7  | 13.0  | 6.4  | (27) |
| (28) |                                                                                  |             | MCD B  | 4.5   | 8.9   | 15.8  | 22.7  | 25.3  | 28.0  | 30.8  | 30.5  | 25.9  | 21.6  | 15.4  | 7.7  | (28) |
| (29) |                                                                                  | 2%          | WB     | 1.1   | 3.1   | 8.9   | 13.8  | 17.7  | 21.6  | 25.6  | 25.8  | 21.2  | 15.9  | 11.1  | 3.8  | (29) |
| (30) |                                                                                  |             | MCD B  | 2.6   | 5.5   | 13.5  | 19.6  | 23.9  | 26.6  | 29.1  | 29.6  | 24.1  | 20.0  | 13.6  | 5.3  | (30) |
| (31) |                                                                                  | 5%          | WB     | -0.1  | 1.5   | 6.7   | 12.2  | 16.7  | 20.7  | 24.8  | 25.0  | 20.2  | 14.9  | 9.6   | 2.1  | (31) |
| (32) |                                                                                  |             | MCD B  | 1.4   | 3.7   | 10.8  | 17.8  | 22.6  | 25.4  | 27.9  | 28.5  | 23.7  | 18.3  | 11.7  | 3.6  | (32) |
| (33) |                                                                                  | 10%         | WB     | -1.3  | 0.3   | 5.0   | 10.7  | 15.7  | 20.0  | 24.2  | 24.3  | 19.2  | 13.8  | 8.1   | 0.7  | (33) |
| (34) |                                                                                  |             | MCD B  | 0.2   | 2.3   | 8.7   | 16.1  | 21.2  | 24.3  | 26.9  | 27.2  | 22.6  | 17.1  | 10.0  | 1.9  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.4  | 11.3  | 10.7  | 12.9  | 13.0  | 10.7  | 7.4   | 7.9   | 11.0  | 12.3  | 9.7   | 10.0 | (35) |
| (36) |                                                                                  |             | MCD BR | 11.1  | 12.6  | 15.3  | 17.4  | 17.4  | 14.8  | 10.4  | 10.2  | 12.8  | 14.7  | 12.4  | 9.8  | (36) |
| (37) |                                                                                  |             | MCWBR  | 8.9   | 9.1   | 9.4   | 8.9   | 7.6   | 6.3   | 4.7   | 5.0   | 6.2   | 8.2   | 8.6   | 8.1  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 9.8   | 11.0  | 14.2  | 14.6  | 14.2  | 10.9  | 7.8   | 8.6   | 9.7   | 12.2  | 10.7  | 9.3  | (38) |
| (39) |                                                                                  |             | MCWBR  | 8.4   | 8.2   | 9.2   | 8.3   | 7.2   | 5.3   | 4.1   | 5.0   | 5.6   | 7.5   | 8.1   | 7.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.392  | 0.463 | 0.531 | 0.602 | 0.614 | 0.685 | 0.648 | 0.554 | 0.485 | 0.501 | 0.460 | 0.398 | (40) |      |
| (41) |                                                                                  | taud        | 1.965  | 1.836 | 1.743 | 1.653 | 1.669 | 1.590 | 1.723 | 1.939 | 2.045 | 1.888 | 1.916 | 1.994 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 726    | 721   | 718   | 699   | 702   | 654   | 674   | 730   | 756   | 677   | 643   | 683   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 131    | 172   | 209   | 243   | 244   | 265   | 230   | 181   | 153   | 161   | 138   | 119   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.48   | 3.30  | 4.27  | 4.89  | 5.48  | 5.09  | 4.04  | 4.24  | 4.15  | 3.41  | 2.29  | 2.08  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.15   | 0.23  | 0.39  | 0.42  | 0.53  | 0.49  | 0.57  | 0.50  | 0.42  | 0.24  | 0.27  | 0.16  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (27 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +100    | +59     |

Nomenclature: See separate page